

AXCEL MUGISHA

M.Sc. Student in Mathematics & Computer Science for AI (IMA)

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Summary

Master's student in Mathematics and Computer Science for AI (IMA) aiming for a Ph.D. in Machine Learning. My research focuses on applied mathematics for ML, at the intersection of optimization, probability theory, and deep learning.

Education

M.Sc. in Mathematics & Computer Science for AI (IMA) **2025–2027**

University of Toulouse, France

B.Sc. in Mathematics (Selective Honors Program / Parcours Spéciaux) **2022–2025**

University of Toulouse, France

Experience

Perception & Control Engineer (Driverless) – TLSE Racing **2025–Present**

Formula Student, Toulouse

- Developed the multi-sensor perception pipeline for an autonomous formula race car (synchronous LiDAR and camera fusion)
- Implemented state estimation (position, velocity, yaw) using an Extended Kalman Filter (EKF)
- Implemented real-time trajectory planning and dynamic path-following algorithms

Research Intern – Detection of diffusion model-generated deepfakes **Apr. 2026–Sept. 2026**

Toulouse Mathematics Institute (IMT), France

- Image reconstruction via diffusion models to extract distinctive artifacts of generated images
- Evaluated multi-domain reconstruction error maps: pixel, latent, wavelet, frequency, and residual noise spaces
- Trained and benchmarked deep learning models to discriminate between real and synthetic images

Machine Learning / Optimization Intern **Mar. 2025–May 2025**

University of Toulouse, France

- Theoretical study of duality conditions, Safe, and Gap Safe screening rules for high-dimensional LASSO
- Algorithmic implementation in Python (ISTA and coordinate descent) and empirical convergence analysis on real and synthetic data

Projects

Obstacle Detection via Fisheye Camera (LAAS-CNRS)

- Geometric calibration of wide-angle cameras using OpenCV and PyTorch CNN implementation for real-time obstacle detection and distance estimation.

Compressed Sensing Reconstruction & AMP Algorithm

- Implementation and benchmarking of the Approximate Message Passing algorithm on low-rank matrices, including theoretical phase transition and noise robustness analysis.

Random Coalescent Processes

- Genealogy modeling via continuous-time Markov chains and Python numerical simulation to analyze asymptotic extinction times.

Self-Similar Data Representation

- Data modeling via attractors of Iterated Function Systems (IFS), combining Hausdorff dimension estimation and parameter space numerical exploration.

Honors & Awards

ANITI Graduate School Excellence Scholarship (2025, 2026): Merit-based academic fellowship awarded by the Interdisciplinary Artificial Intelligence Institute of Toulouse.

Skills

Languages & Tools	Python (PyTorch, Scikit-Learn, NumPy, SciPy, Matplotlib), MATLAB, C++, Git, Linux, L ^A T _E X
Applied Mathematics	Stochastic Modeling, Convex Optimization, Extended Kalman Filtering (EKF), Bayesian Inference
Languages	French: Bilingual / Native English: Fluent Kirundi: Native language